Name : Adhithya.S
Reg. No: 192412352

CODE:

```
# House Price Prediction using Linear Regression

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample House Dataset
data = {
    "Area": [
        800, 1000, 1200, 1500, 1800,
        2000, 2200, 2500, 2800, 3000
    ],

    "Bedrooms": [
        1, 2, 2, 3, 3,
        3, 4, 4, 4, 5
    ],

    "Bathrooms": [
        1, 1, 2, 2, 2,
        3, 3, 3, 4, 4
    ],

    "Price": [
        2500000, 3200000, 4000000, 5500000, 6500000,
        7500000, 8500000, 10000000, 11500000, 13000000
    ]
}

# Convert dataset into DataFrame
df = pd.DataFrame(data)

print("HOUSE PRICE DATASET")
print("=" * 50)
print(df)

# Input Features
X = df[["Area", "Bedrooms", "Bathrooms"]]

# Target Variable
y = df["Price"]

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create Linear Regression Model
model = LinearRegression()

# Train the Model
model.fit(X_train, y_train)
```

```
# Predict Test Data
y_pred = model.predict(X_test)

# Calculate Performance
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

# Display Results
print("\nMODEL PERFORMANCE")
print("=" * 50)

print("Mean Squared Error:", round(mse, 2))
print("R-squared Score:", round(r2, 4))

print("\nActual Prices:", list(y_test))
print("Predicted Prices:",
      [round(price, 2) for price in y_pred])

# Predict Price for a New House
new_house = pd.DataFrame({
    "Area": [2000],
    "Bedrooms": [3],
    "Bathrooms": [2]
})

predicted_price = model.predict(new_house)

print("\nNEW HOUSE DETAILS")
print("=" * 50)

print("Area:", new_house["Area"].iloc[0], "sq.ft")
print("Bedrooms:", new_house["Bedrooms"].iloc[0])
print("Bathrooms:", new_house["Bathrooms"].iloc[0])

print("\nPredicted House Price: ₹",
      round(predicted_price[0], 2))
```

OUTPUT :

```
HOUSE PRICE DATASET
==================================================
   Area  Bedrooms  Bathrooms     Price
0   800         1          1   2500000
1  1000         2          1   3200000
2  1200         2          2   4000000
3  1500         3          2   5500000
4  1800         3          2   6500000
5  2000         3          3   7500000
6  2200         4          3   8500000
7  2500         4          3  10000000
8  2800         4          4  11500000
9  3000         5          4  13000000

MODEL PERFORMANCE
==================================================
Mean Squared Error: 25224929127.07
R-squared Score: 0.9985

Actual Prices: [11500000, 3200000]
Predicted Prices: [np.float64(11619843.92), np.float64(3010033.44)
]

NEW HOUSE DETAILS
==================================================
Area: 2000 sq.ft
Bedrooms: 3
Bathrooms: 2

Predicted House Price: ₹ 7748606.47
```